

Wrangling

2026-07-17

Initial Wrangling

```
# Read in data
## Goal data
hockey <- read_csv("data/hockey.txt")
```

New names:

Rows: 8086 Columns: 18

-- Column specification

```
----- Delimiter: "," chr
(14): Tm, ...4, Opp, ...6, ...7, Scorer, Age, Assist1, Assist2, Goalie,... dbl
(2): Rank, Cr. Goal # date (1): Date time (1): Time
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...4`
* `` -> `...6`
* `` -> `...7`
```

```
## Player data
players <- read_csv("data/player_data.txt")
```

Rows: 1206 Columns: 32

-- Column specification -----

Delimiter: ","

chr (6): Player, Team, Pos, TOI, Awards, -9999

dbl (25): Rk, Age, GP, G, A, PTS, +/-, PIM, EVG, PPG, SHG, GWG, EV, PP, SH,...

time (1): ATOI

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# Basic wrangling
hockey_clean <- hockey %>%
  ## Select columns of interest
  select(Tm, Assist1, Scorer) %>%
  ## Drop NA (get rid of unassisted goals)
  drop_na()

# Wrangle player data
players <- players %>%
  select(Player, G, A, Pos, , GP, Team) %>%
  rename(name = Player, ## For joining
         "Total Goals" = G,
         "Total Assists" = A,
         "Games Played" = GP,
         "Position" = Pos) %>%
  # Rename Position variable levels
  mutate(Position = case_when(Position == "C" ~ "Center",
                              Position == "RW" ~ "Right Wing",
                              Position == "LW" ~ "Left Wing",
                              Position == "W" ~ "Wing",
                              Position == "F" ~ "Forward",
                              Position == "D" ~ "Defense",
                              TRUE ~ "Goalie"))

## Save for Shiny app
saveRDS(players, "app/hockey_networks/players.rds")
```

Shiny App Wrangling

```
# For reproducibility
set.seed(231)

# Using hockey_clean as basis

# SJS
hockey_t <- hockey_clean %>%
  filter(Tm == "SJS") %>%
  # Count total of goals between a pair
```

```

    group_by(Assist1, Scorer) %>% ## Groups by pair
    summarize(goals = n()) ## Counts number of goals between the pairs
# Make directional network
hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

# Get edge weights for HITS scores
hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

# Compute HITS scores
hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

# Add HITS scores as vertex attributes
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

# Make plottable on ggplot
hockey_network <- ggnetwork(hockey_igraph)

# Save for Shiny app
saveRDS(hockey_network, "app/hockey_networks/sjs_network.rds")

# MTL
hockey_t <- hockey_clean %>%
  filter(Tm == "MTL") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/mtl_network.rds")

# CAR
hockey_t <- hockey_clean %>%
  filter(Tm == "CAR") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/car_network.rds")

# EDM
hockey_t <- hockey_clean %>%
  filter(Tm == "EDM") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/edm_network.rds")

# TBL
hockey_t <- hockey_clean %>%
  filter(Tm == "TBL") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/tbl_network.rds")

# BUF
hockey_t <- hockey_clean %>%
  filter(Tm == "BUF") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/buf_network.rds")

# CBJ
hockey_t <- hockey_clean %>%
  filter(Tm == "CBJ") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/cbj_network.rds")

# PIT
hockey_t <- hockey_clean %>%
  filter(Tm == "PIT") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/pit_network.rds")

# PIT
hockey_t <- hockey_clean %>%
  filter(Tm == "PHI") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/phi_network.rds")

# PIT
hockey_t <- hockey_clean %>%
  filter(Tm == "ANA") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/ana_network.rds")

# BOS
hockey_t <- hockey_clean %>%
  filter(Tm == "BOS") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/bos_network.rds")

# VEG
hockey_t <- hockey_clean %>%
  filter(Tm == "VEG") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```



```

saveRDS(hockey_network, "app/hockey_networks/veg_network.rds")

# UTA
hockey_t <- hockey_clean %>%
  filter(Tm == "UTA") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/uta_network.rds")

# COL
hockey_t <- hockey_clean %>%
  filter(Tm == "COL") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/col_network.rds")

# NSH
hockey_t <- hockey_clean %>%
  filter(Tm == "NSH") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/nsh_network.rds")

# VAN
hockey_t <- hockey_clean %>%
  filter(Tm == "VAN") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/van_network.rds")

# WSH
hockey_t <- hockey_clean %>%
  filter(Tm == "WSH") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/wsh_network.rds")

# NYI
hockey_t <- hockey_clean %>%
  filter(Tm == "NYI") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/nyi_network.rds")

# NJD
hockey_t <- hockey_clean %>%
  filter(Tm == "NJD") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/njd_network.rds")

# NYR
hockey_t <- hockey_clean %>%
  filter(Tm == "NYR") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/nyr_network.rds")

# MIN
hockey_t <- hockey_clean %>%
  filter(Tm == "MIN") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/min_network.rds")

# LAK
hockey_t <- hockey_clean %>%
  filter(Tm == "LAK") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/lak_network.rds")

# CGY
hockey_t <- hockey_clean %>%
  filter(Tm == "CGY") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/cgy_network.rds")

# CHI
hockey_t <- hockey_clean %>%
  filter(Tm == "CHI") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/chi_network.rds")

# DAL
hockey_t <- hockey_clean %>%
  filter(Tm == "DAL") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/dal_network.rds")

# DET
hockey_t <- hockey_clean %>%
  filter(Tm == "DET") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/det_network.rds")

# FLA
hockey_t <- hockey_clean %>%
  filter(Tm == "FLA") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/fla_network.rds")

# OTT
hockey_t <- hockey_clean %>%
  filter(Tm == "OTT") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```



```

saveRDS(hockey_network, "app/hockey_networks/ott_network.rds")

# SEA
hockey_t <- hockey_clean %>%
  filter(Tm == "SEA") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/sea_network.rds")

# STL
hockey_t <- hockey_clean %>%
  filter(Tm == "STL") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```

saveRDS(hockey_network, "app/hockey_networks/stl_network.rds")

# TOR
hockey_t <- hockey_clean %>%
  filter(Tm == "TOR") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

saveRDS(hockey_network, "app/hockey_networks/tor_network.rds")

# WPG
hockey_t <- hockey_clean %>%
  filter(Tm == "WPG") %>%
  group_by(Assist1, Scorer) %>%
  summarize(goals = n())

hockey_igraph <- graph_from_data_frame(hockey_t, directed = TRUE)

hockey_edge_weights <- edge_attr(hockey_igraph, name = "goals")

hockey_hits <- hits_scores(hockey_igraph, weights = hockey_edge_weights)

hockey_igraph <- set_vertex_attr(hockey_igraph, name = "hub_score",
                                value = hockey_hits$hub)
hockey_igraph <- set_vertex_attr(hockey_igraph, name = "auth_score",
                                value = hockey_hits$authority)

hockey_network <- ggnetwork(hockey_igraph)

```

```
saveRDS(hockey_network, "app/hockey_networks/wpg_network.rds")
```

Data

2025-26 NHL Skater Statistics. (2026). [Dataset]. Sports Reference. Retrieved May 1, 2026, from https://www.hockey-reference.com/leagues/NHL_2026_skaters.html

Goal Finder, Current Search for all games, in 2025-26, in the regular season, sorted by descending Scorer. (2026). [Dataset]. Sports Reference. Retrieved April 15 & 17, 2026, from <https://www.sports-reference.com/stathead/tiny/FtM4c>